

Generative AI for Concept Creation in Footwear Design

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Figure 1: AI driven concept to prototype process: (1) web interface of AI ARCHIVE, (2) selected AI-generated concept, (3) generated 3D model, (4) and final physical prototype.

ABSTRACT

We present AI ARCHIVE, a footwear design tool using generative artificial intelligence (AI) that we successfully integrated into the design process at adidas. AI ARCHIVE is based on diffusion models and was trained on the entire archive of adidas sneakers, which dates back to the company's beginning in the 1950ies. Being trained on this unique dataset enables the AI to generate new and innovative sneaker designs that draw inspiration from the archive and pay homage to the rich history of the adidas brand. AI ARCHIVE has been rolled out to our designers as a web application in 2022. The tool has since established itself as an essential ingredient in the concept-to-prototype process of many of our designers.

The proposed system gives users a high level of control over the design process, enabling them to precisely guide the AI to create designs according to their direction.

We believe that the use of generative AI in footwear design has the potential to transform the industry, as it allows designers to explore hundreds of different concepts in almost no time.

KEYWORDS

Footwear, Concept Design, AI, Generative AI, Diffusion Models

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1 INTRODUCTION

Footwear design is a complex process that involves multiple stages, including ideation, prototyping, and production. However, perhaps the most critical stage of the process is the inspiration phase, where designers draw on a range of sources to develop creative ideas for their designs. This stage is essential to the success of a footwear design, as it sets the direction and tone for the entire design process.

Inspiration can come from a variety of sources, including fashion trends, history, cultural influences, and even art, architecture or other related disciplines. It is up to the designer to interpret and integrate these sources of inspiration into their designs, creating footwear that is both aesthetically pleasing and functional.

Having collected the inspirational material, designers start ideating and sketching initial concepts of the shoe. These sketches are typically rough and conceptual, with designers exploring different ideas and directions for the design. At this stage, designers may create dozens or even hundreds of sketches and mockups, each exploring a different aspect of the design.

Generative AI has recently emerged as a powerful tool for generating novel and unique content. It has been successfully used in a wide range of applications, including magazine covers, advertisements, and marketing campaigns. Tools like Dall-E 2, Stable Diffusion, and Midjourney took the internet by storm and have been adopted by millions of users within less than one year of existence. What makes the use of generative AI appealing for the design concept phase is its ability to create a huge variety of concepts in a short amount of time. To do this successfully, the AI needs to provide exactly the right amount of control to the designers, such that they can guide it to generate concepts according to their direction while still being an intuitive and fun tool to use.

2 GENERATIVE AI FOR SNEAKER DESIGN

To support the creative design process in adidas, we created AI ARCHIVE, a generative AI based on a custom-trained diffusion

model[Rombach et al. 2022]. Diffusion models are a framework for generative AI capable of generating diverse and high-quality images. They are based on neural networks trained to iteratively remove noise from images, starting from pure noise and successively restoring the underlying image.

An important benefit of diffusion models is that they are well-suited for controlled generation of images. For our work presented here, we conditionally train a model to use CLIP [Radford et al. 2021] embeddings of our training images in its denoising process. CLIP is a large, pre-trained neural network that turns images into highly compressed embedding vectors that encode their content and style. During inference, we can use these CLIP embeddings to steer our model toward making images similar to the embedded image. The abstract nature of the embeddings also allows users to combine multiple embeddings to create new embeddings that lies somewhere in between the existing vectors. When done correctly, the diffusion model will then produce a fusion between the embedded images. This means that users can mix and match features of different shoes and mood images to synthesize interesting new designs.

Moreover, we can use diffusion models to make edits to an existing image using a process called inpainting. Every timestep during image denoising, we replace a part of the noisy image with a preset part of an existing image. This forces the diffusion model to fill in the noisy gaps while trying to keep the entire image consistent. This allows users to synthesize targeted edits of existing images. For example, designers can use inpainting to generate new uppers on top of an existing midsole or edit the laces of a shoe (see Figure 2).

We trained AI ARCHIVE on our internal database of shoe images. This database contains over 160,000 lateral views of adidas shoes, providing a wealth of data for the AI to learn from. By training on this extensive archive, the AI was able to develop a deep understanding of the design principles and patterns that have characterized adidas footwear throughout the decades. This allows the AI to create new and innovative footwear concepts that pay homage to the rich history of the adidas brand.

AI ARCHIVE was realized as a cloud-native web application and made accessible to all employees at adidas. The web interface allows users to upload their inspirational content, either in form of images or text prompts, and to assign weights to each input. Image synthesis then happens in the cloud and the results are typically fed

back to the user within less than 30 seconds. The user can further refine the generated designs by tweaking input content, weights, parameters, and random seeds.

3 CONCEPT TO PROTOTYPE

After the ideation process, designers and designated product creation experts move on to the prototyping stage. This typically starts with handing over the design to 3D creation for further development. The first step in 3D creation of a digital model is placing the digital shoe last and overlaying the AI-generated concepts. From here 3D modeling and sculpting are used to create a digital representation of the shoe design, including all the relevant details such as the shape, volume, and individual parts. For specific designs, such as the outsole pattern, the creation of parametric-design tools is considered to explore different design options and ensure reproducibility. Once the geometry is finished, digital materialization is key to achieving a photorealistic rendering of the design. This involves texture scanning and creation, shading, and material application. The final rendering helps to validate the idea and concept and allows to make necessary changes before moving on to physical prototyping.

The physical prototyping process involves several stages, including pattern making, upper and midsole creation, and assembly. The upper is created using a variety of techniques, such as cutting, stitching, and bonding, to achieve the desired shape and fit. The midsole is already created through the 3D creation process and can be prototyped with additive manufacturing solutions such as 3D printing. Once the upper and midsole are complete, the shoe is assembled by combining the upper with the sole and any other components. This is typically done by skilled craftsmen or using specialized machinery.

4 OUTLOOK

We presented AI ARCHIVE, a custom-trained diffusion model that we successfully integrated into the creative process at adidas. The tool was well adopted by our design community and is now used across all business units of the company. The success of the tool was driven primarily by three factors: (1) Making it accessible via a web interface significantly reduced the entry burden and invites people to start playing with the tool; (2) Having an internal tool trained exclusively on adidas data results in a clear legal situation (3) Giving a maximum of control to the users greatly improved the acceptance of the tool, as it enables them to precisely guide the AI into a desired direction.

We are currently exploring automatically generating the first 3D models directly from our diffusion model. This will further streamline the creation process and remove manual labor when going from design to production, and also enable the use of AI-generated content directly in video games or metaverse applications.

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Figure 2: Inpainting (left) allows the user to generate variations of selected parts only. Sneaker variation creates new designs from existing sneakers and mood images (right)